

Samsung Innovoation Campus(SIC)-IoT Internship Report

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by*

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ABSTRACT

With the rapid increase in vehicle population and urbanization, efficient parking management has become a significant challenge. Traditional parking systems lack real-time monitoring and automation, leading to traffic congestion, fuel wastage, and inefficient utilization of parking spaces.

This project presents an IoT-based Smart Parking System designed to provide real-time information on parking slot availability. The system utilizes ultrasonic sensors to detect vehicle entry, exit, and slot occupancy, while an ESP32 microcontroller processes the data and communicates using the MQTT protocol. A Node-RED dashboard is used for real-time visualization, displaying parking status, available slots, and environmental parameters such as temperature and humidity.

The system also incorporates an event-driven communication mechanism to ensure efficient data transmission and reduced network overhead. Experimental results demonstrate that the system provides accurate, reliable, and low-latency performance.

Overall, the proposed system offers a cost-effective, scalable, and efficient solution for modern parking management and can be extended for smart city applications.

Furthermore, the system provides visual feedback through LED indicators, enabling users to quickly understand parking availability without relying solely on digital interfaces. The integration of environmental monitoring using the DHT11 sensor adds an additional layer of functionality, allowing the system to track temperature and humidity conditions within the parking area. This feature can be extended for safety monitoring and ventilation control in enclosed parking environments.

The modular design of the system ensures flexibility and ease of scalability. Additional parking slots can be incorporated by extending sensor nodes and updating MQTT topics without altering the core system architecture. The use of open-source tools such as Arduino IDE and Node-RED further enhances adaptability.

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Chapter 1

Introduction to Internet of Things

1.1 Introduction

The Internet of Things (IoT) refers to the interconnection of physical devices embedded with sensors, software, and communication technologies that enable them to collect and exchange data over the internet. IoT has emerged as a transformative technology that enables automation, real-time monitoring, and intelligent decision-making across various domains.

In recent years, IoT has gained significant importance due to the rapid growth of smart devices, wireless communication, and cloud computing. It plays a crucial role in developing smart systems that enhance efficiency, reduce human effort, and improve overall system performance.

1.2 Characteristics of IoT

IoT systems exhibit several important characteristics:

- **Connectivity:** Devices are interconnected through wired or wireless networks.
- **Real-time Data Processing:** Continuous monitoring and instant data updates.
- **Automation:** Systems operate with minimal human intervention.
- **Scalability:** Ability to expand the system by adding more devices.
- **Intelligence:** Data-driven decision making using sensors and analytics.

1.3 Applications of IoT

IoT technology is widely used in various fields:

- Smart Cities (traffic management, smart parking)
- Healthcare Monitoring Systems
- Industrial Automation
- Smart Homes
- Environmental Monitoring

Among these, smart parking systems have become one of the most important applications in urban environments due to increasing vehicle density and parking challenges.

1.4 IoT in Smart Parking Systems

Smart parking systems utilize IoT technology to monitor parking spaces in real time and provide accurate information about slot availability. Sensors are used to detect vehicle presence, while microcontrollers process the data and communicate with cloud platforms or dashboards.

The integration of IoT in parking systems helps in reducing traffic congestion, minimizing fuel consumption, and improving user convenience. Real-time data visualization and alert mechanisms further enhance the efficiency of parking management systems.

1.5 Components of IoT System

A typical IoT system consists of the following components:

- **Sensors:** Collect real-world data such as distance, temperature, and humidity.
- **Microcontroller:** Processes sensor data (e.g., ESP32).
- **Communication Module:** Enables data transmission using protocols like Wi-Fi and MQTT.

- **Cloud/Server:** Stores and processes data.
- **User Interface:** Displays data through dashboards (e.g., Node-RED).

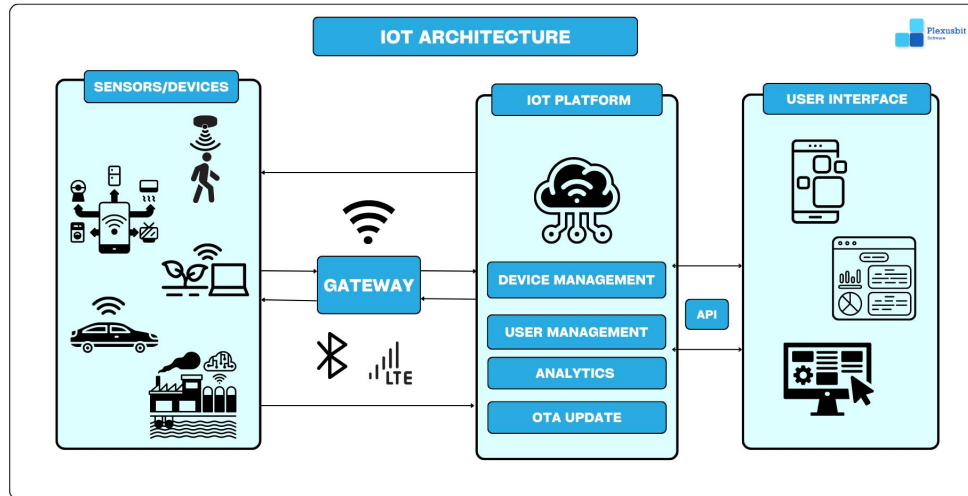


FIGURE 1.1: IOT Architecture

1.6 Overview of Proposed Work

During the internship, an IoT-based Smart Parking System was developed using ESP32, ultrasonic sensors, MQTT protocol, and Node-RED dashboard. The system provides real-time information about parking slot availability, vehicle entry and exit tracking, and environmental monitoring.

The project demonstrates the practical implementation of IoT concepts in solving real-world problems related to parking management.

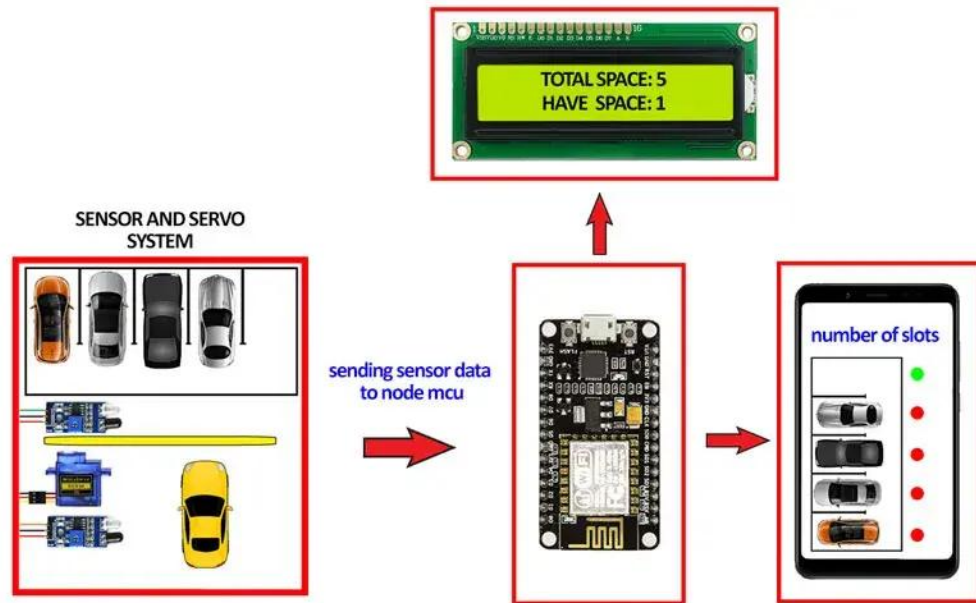


FIGURE 1.2: Smart Parking System Overview

1.7 Organization of the Report

The report is organized as follows:

- **Chapter 1** introduces the fundamentals of IoT and its applications.
- **Chapter 2** describes the practical experiments and learning during the internship.
- **Chapter 3** explains the Smart Parking System project in detail.
- **Chapter 4** presents the implementation and working of the system.
- **Chapter 5** discusses the results and analysis.
- **Chapter 6** provides the conclusion.
- **Chapter 7** outlines future scope and enhancements.

Chapter 2

Practical Experiments and Learning

2.1 Introduction

During the internship, several practical experiments were performed to understand the fundamentals of IoT systems. These experiments focused on sensor interfacing, microcontroller programming, communication protocols, and real-time data visualization. The knowledge gained from these experiments formed the foundation for the development of the Smart Parking System.

2.2 Experiment 1: Interfacing Ultrasonic Sensor with ESP32

The ultrasonic sensor (HC-SR04) is used to measure distance by transmitting ultrasonic waves and receiving the reflected signal. It plays a key role in detecting vehicle presence in parking systems.

Objective: To measure distance using an ultrasonic sensor and display the output.

Components Used:

- ESP32 Microcontroller

- HC-SR04 Ultrasonic Sensor
- Connecting wires

Working Principle: The sensor emits ultrasonic waves and calculates distance based on the time taken for the echo to return.

Observation: Accurate distance measurement was achieved within the range of 2 cm to 400 cm.

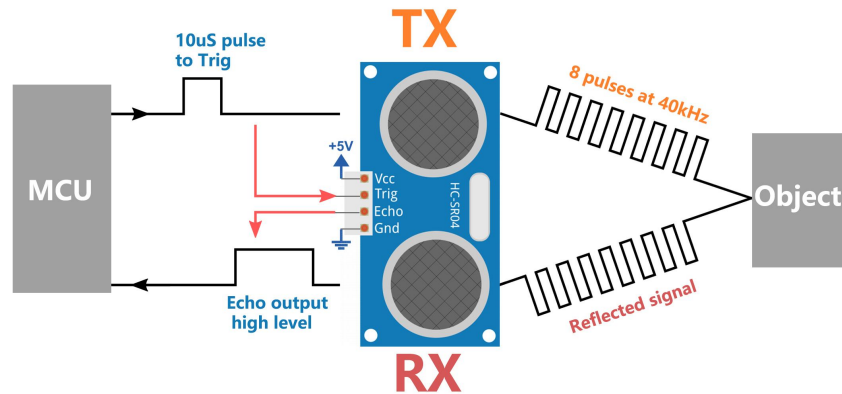


FIGURE 2.1: Working Principle of Ultrasonic Sensor

2.3 Experiment 2: Interfacing DHT11 Sensor

The DHT11 sensor is used for measuring temperature and humidity.

Objective: To read environmental parameters using ESP32.

Components Used:

- ESP32 Microcontroller
- DHT11 Sensor

Working Principle: The sensor uses a digital signal to transmit temperature and humidity values to the microcontroller.

Observation: Temperature and humidity values were successfully monitored and displayed in real time.

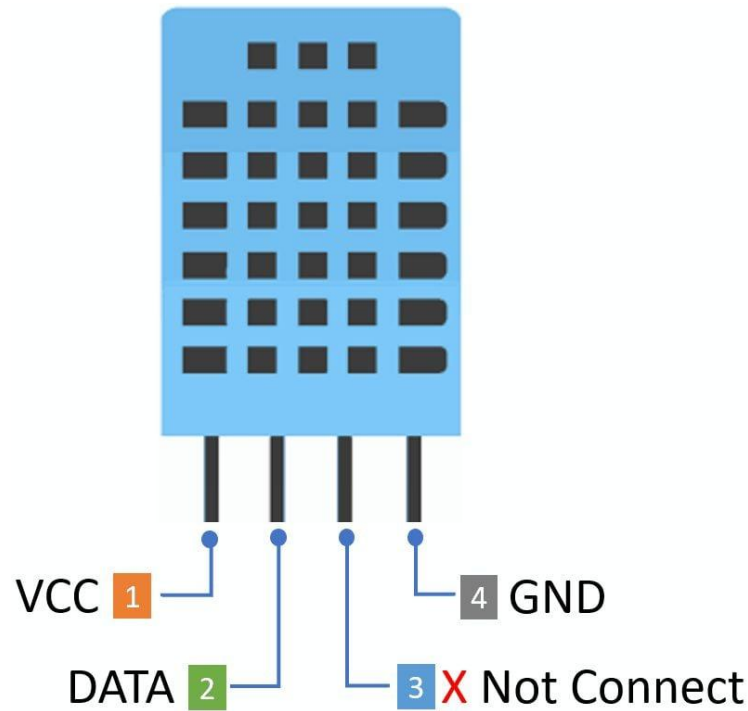


FIGURE 2.2: DHT11 Temperature and Humidity Sensor

2.4 Experiment 3: Wi-Fi Connectivity using ESP32

The ESP32 has built-in Wi-Fi capability, which enables communication with cloud platforms.

Objective: To connect ESP32 to a Wi-Fi network and verify connectivity.

Working Principle: The ESP32 connects to a Wi-Fi network using SSID and password, enabling internet-based communication.

Observation: Stable Wi-Fi connection was established, allowing data transmission over the internet.

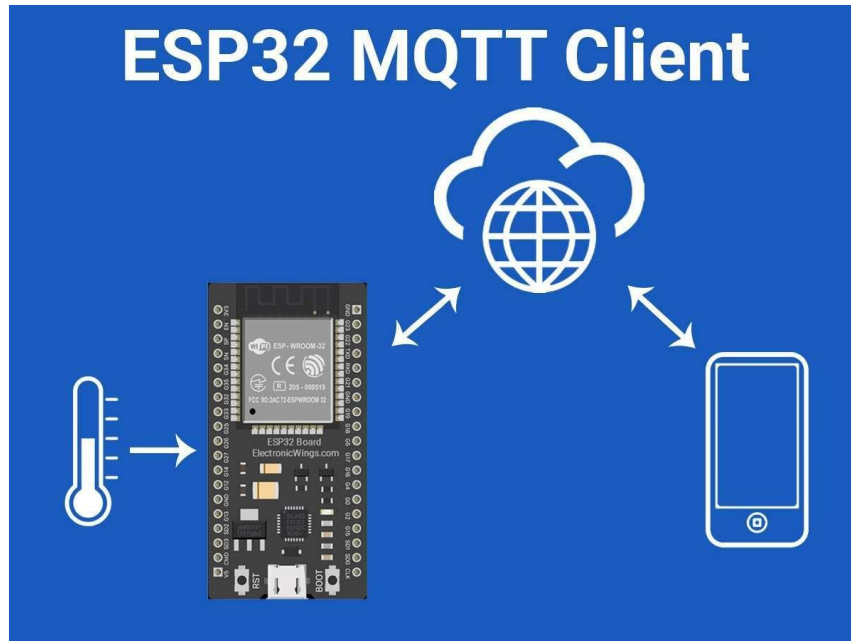


FIGURE 2.3: ESP32 Wi-Fi Connectivity

2.5 Experiment 4: MQTT Communication

MQTT is a lightweight communication protocol used in IoT systems.

Objective: To send and receive data using MQTT protocol.

Working Principle: The ESP32 acts as a publisher and sends data to an MQTT broker, while clients such as Node-RED subscribe to the data.

Observation: Efficient real-time communication was achieved with minimal latency.

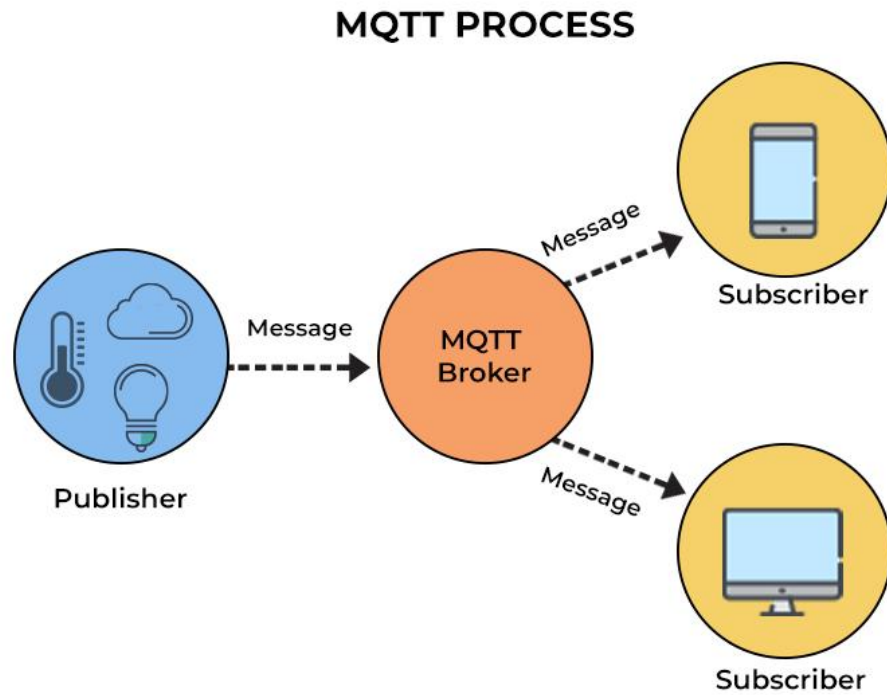


FIGURE 2.4: MQTT Publish-Subscribe Communication Model

2.6 Experiment 5: Node-RED Dashboard

Node-RED is used to visualize IoT data using a graphical interface.

Objective: To display real-time sensor data on a dashboard.

Working Principle: Node-RED subscribes to MQTT topics and displays the data using UI components such as gauges and text fields.

Observation: Real-time monitoring dashboard was successfully created, displaying sensor data clearly.

2.7 Learning Outcomes

The following key concepts were learned during the internship:

- Interfacing sensors with ESP32

-
- Real-time data acquisition
 - Wireless communication using Wi-Fi
 - MQTT protocol for IoT communication
 - Data visualization using Node-RED
 - System integration of hardware and software

These practical experiments provided hands-on experience and helped in understanding the complete workflow of IoT systems, which was later applied in the Smart Parking System project.

Chapter 3

Smart Parking System

3.1 Introduction

The Smart Parking System is an IoT-based solution designed to provide real-time information about parking slot availability. The system integrates sensors, a microcontroller, communication protocols, and a visualization platform to ensure efficient parking management.

The system aims to reduce traffic congestion, minimize fuel consumption, and improve user convenience by guiding drivers directly to available parking spaces.

3.2 Problem Statement

In conventional parking systems, drivers do not have access to real-time information about available parking spaces. This leads to unnecessary searching, increased congestion, and inefficient utilization of parking areas.

Manual systems are not scalable and are prone to errors. Therefore, an automated and intelligent system is required to monitor parking spaces and provide real-time updates.

3.3 Objectives

The main objectives of the Smart Parking System are:

- To detect vehicle entry and exit automatically
- To monitor individual parking slot occupancy
- To provide real-time parking status
- To transmit data using MQTT protocol
- To visualize data using Node-RED dashboard
- To send alerts when parking is full

3.4 Proposed System

The proposed system uses ultrasonic sensors to detect vehicle presence and an ESP32 microcontroller to process the data. The processed data is transmitted to an MQTT broker and visualized using a Node-RED dashboard.

The system also includes LED indicators and an alert mechanism for better user interaction.

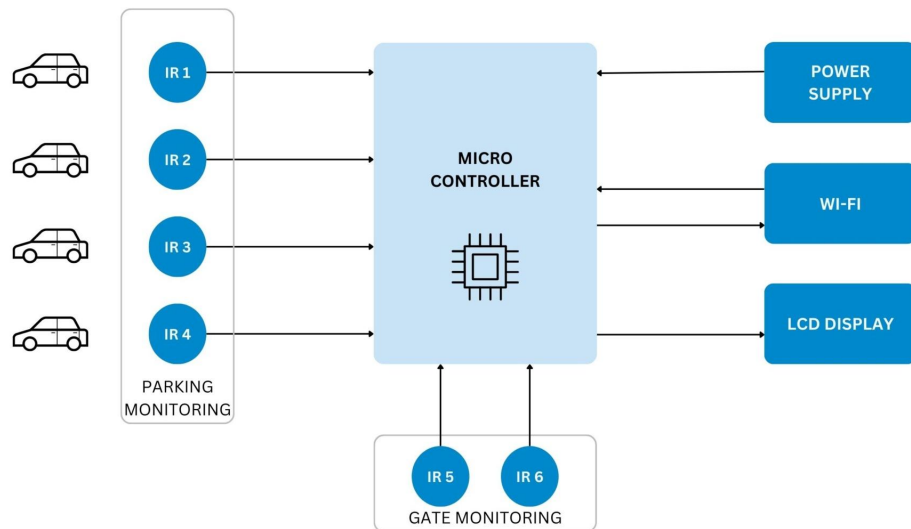


FIGURE 3.1: Block Diagram of Smart Parking System

3.5 System Architecture

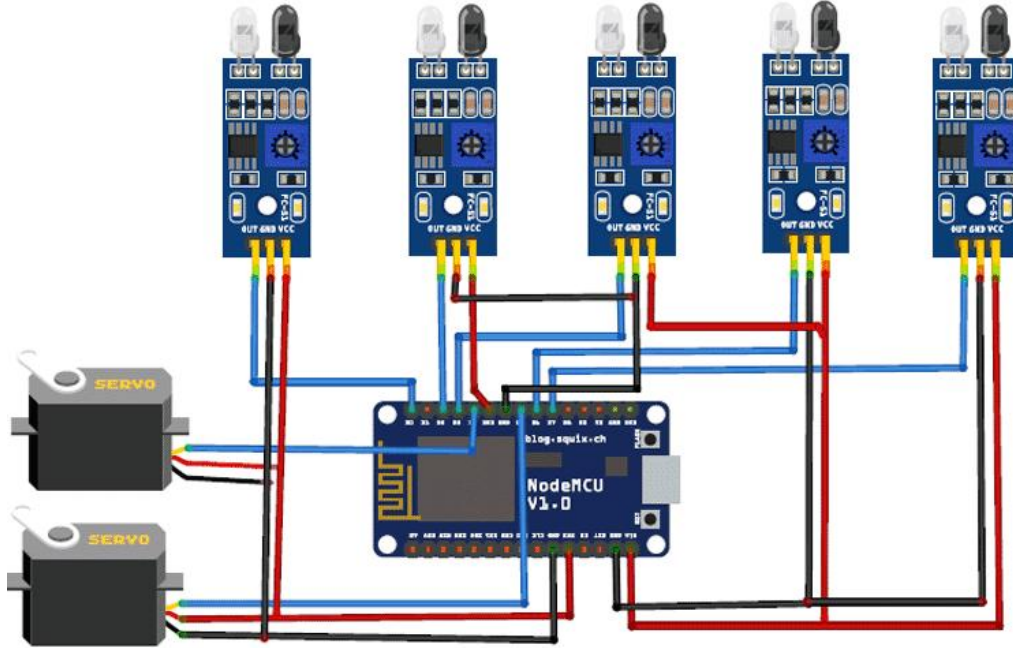


FIGURE 3.2: Smart Parking System Architecture

The system consists of the following layers:

- Sensor Layer (Ultrasonic and DHT11 sensors)
- Processing Layer (ESP32 microcontroller)
- Communication Layer (MQTT protocol)
- Visualization Layer (Node-RED dashboard)

The Smart Parking System follows a layered architecture that ensures efficient data flow, modularity, and scalability. At the sensing layer, ultrasonic sensors are deployed at the entry, exit, and individual parking slots to detect vehicle movement and occupancy status. These sensors continuously measure distance and provide real-time data to the ESP32 microcontroller. Additionally, the DHT11 sensor collects environmental parameters such as temperature and humidity, enhancing system awareness and enabling future extensions like environmental monitoring and safety analysis.

At the processing and communication layers, the ESP32 acts as the central control unit, where sensor data is processed and converted into meaningful information such as vehicle

count, slot availability, and parking status. This processed data is then transmitted using the MQTT protocol, which follows a lightweight publish–subscribe model. The MQTT broker acts as an intermediary between the ESP32 and the visualization platform. Node-RED subscribes to the relevant topics and displays the information on a real-time dashboard. This architecture ensures low latency, efficient communication, and scalability, allowing the system to be easily expanded for larger parking infrastructures or integrated into smart city ecosystems.

3.6 Working Principle

When a vehicle enters the parking area, the entry sensor detects it and increments the vehicle count. Similarly, the exit sensor detects vehicles leaving and decrements the count.

Each parking slot is monitored using an ultrasonic sensor that determines whether the slot is occupied or free based on distance measurement.

The ESP32 processes the sensor data and publishes it to an MQTT broker. Node-RED subscribes to the data and displays it on a dashboard.

In addition to vehicle counting, the system continuously monitors individual parking slots using ultrasonic sensors. Each slot sensor measures the distance between the sensor and the ground. When a vehicle occupies the slot, the measured distance decreases below a predefined threshold, indicating that the slot is occupied. When the vehicle leaves, the distance increases back to its original value, marking the slot as available. This approach ensures accurate and real-time detection of parking occupancy without requiring manual intervention.

To maintain efficient communication and avoid unnecessary data transmission, the system updates the MQTT broker only when there is a change in vehicle count or slot status. This event-driven mechanism reduces network load and improves system performance. The published data includes parking status, number of filled and available slots, environmental parameters, and slot mapping information. Node-RED processes this data to update the dashboard dynamically and trigger alerts when required, such as when the parking lot reaches full capacity.

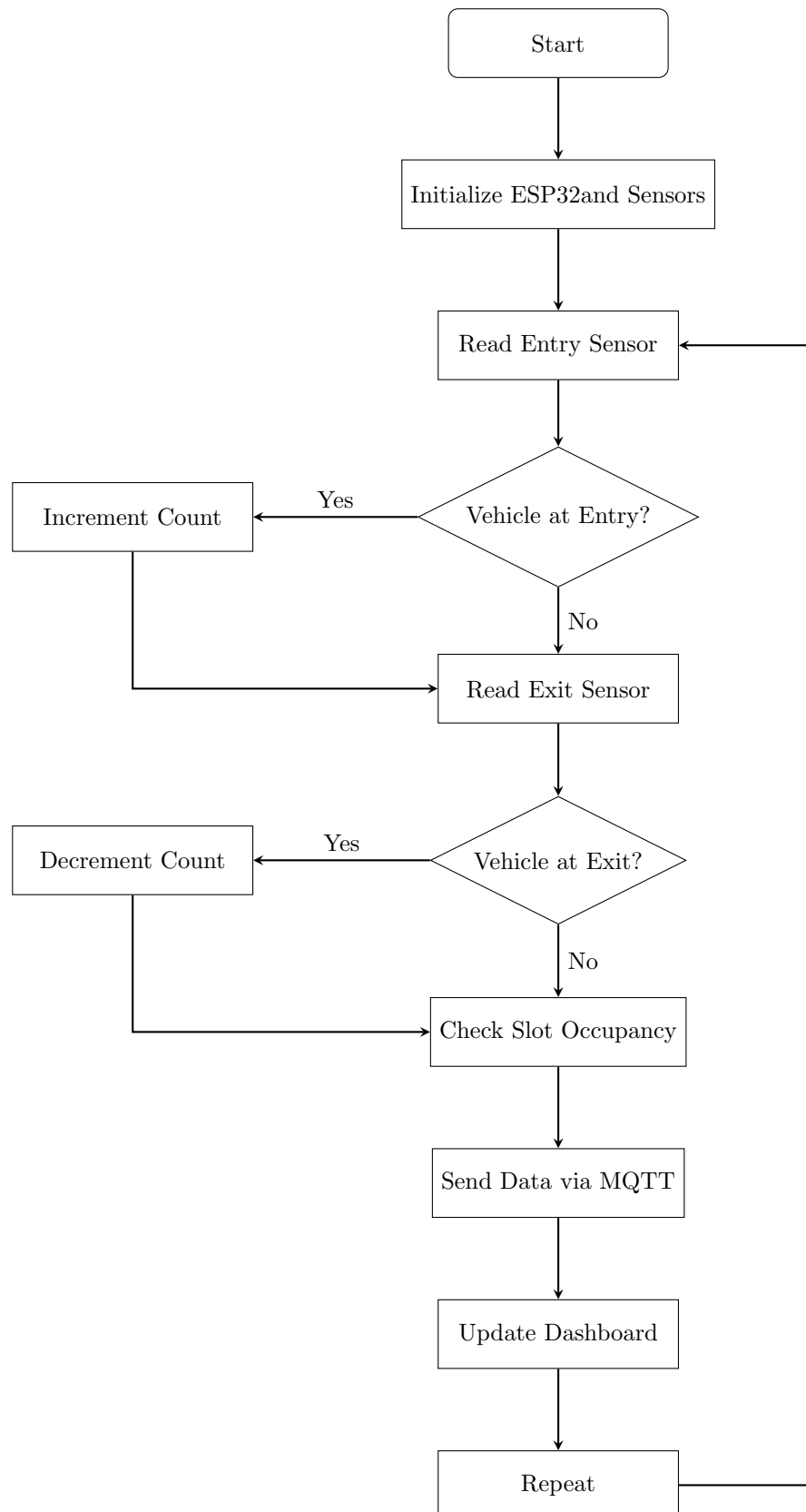


FIGURE 3.3: Flowchart of Smart Parking System

3.7 System Features

- Real-time parking monitoring
- Automatic vehicle counting
- Remote access through dashboard
- Environmental monitoring (temperature and humidity)
- Email alert system for full parking

Chapter 4

Implementation

4.1 Introduction

This chapter describes the implementation of the Smart Parking System, including hardware setup, software development, communication protocols, and system integration. The system is implemented using ESP32, ultrasonic sensors, MQTT protocol, and Node-RED dashboard.

4.2 Hardware Implementation

The hardware setup consists of ultrasonic sensors placed at entry, exit, and parking slots to detect vehicle movement and occupancy. The ESP32 microcontroller is used as the central processing unit.

Components Used:

- ESP32 Microcontroller
- Ultrasonic Sensors (HC-SR04)
- DHT11 Sensor
- LEDs for status indication

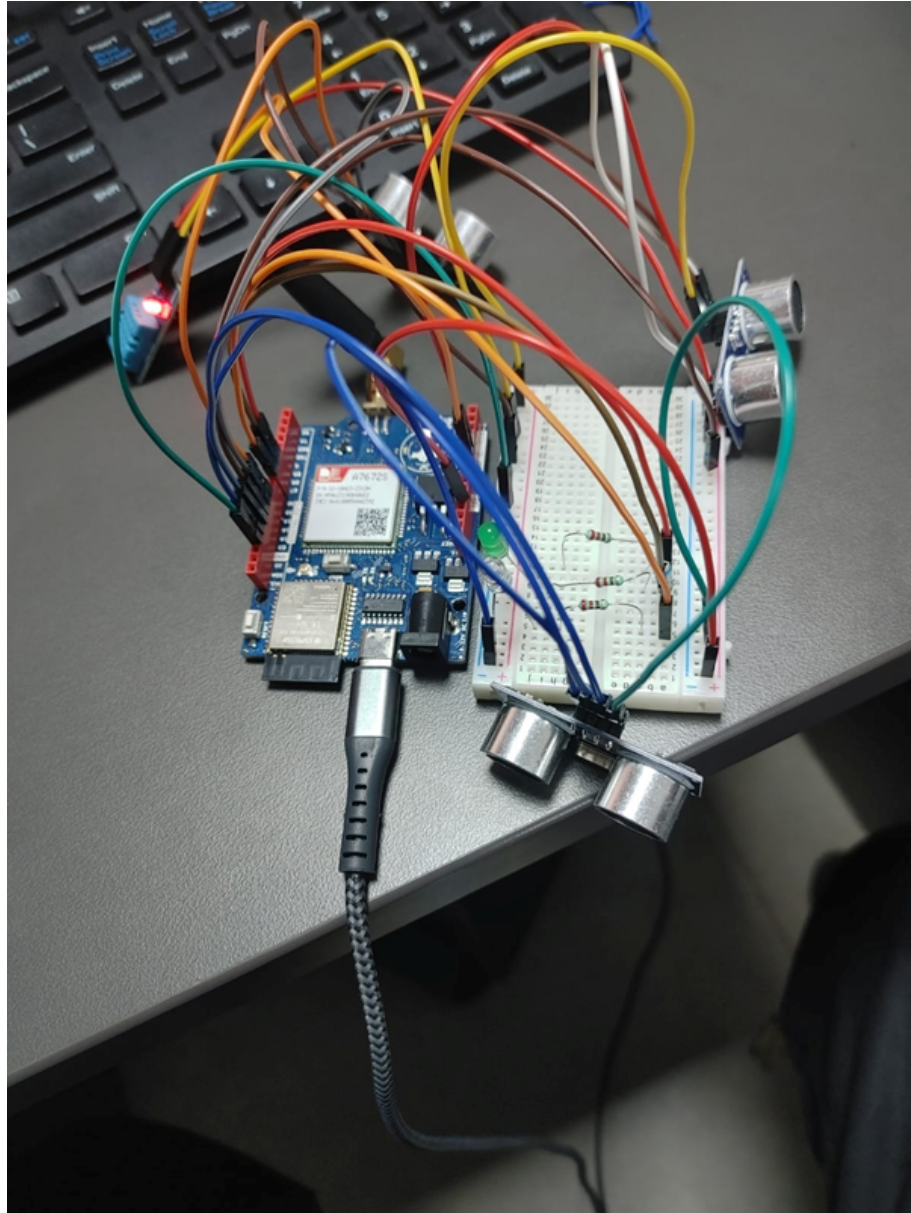


FIGURE 4.1: Hardware Setup of Smart Parking System

The sensors are connected to the ESP32 GPIO pins, and the system is powered using a regulated power supply.

4.3 Software Implementation

The software implementation is carried out using Arduino IDE. The ESP32 is programmed using Embedded C/C++ to read sensor data, process it, and transmit it via MQTT.

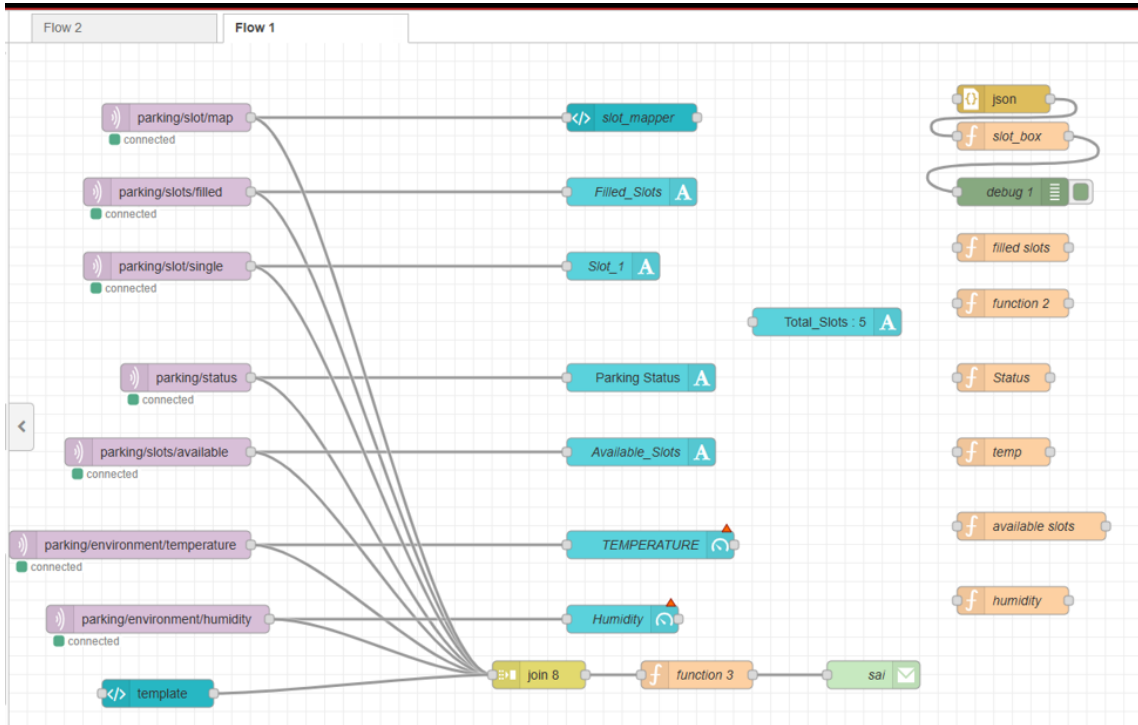


FIGURE 4.2: Node-RED Flow for Smart Parking System

4.3.1 Ultrasonic Sensor Logic

The distance is calculated based on the time taken for the ultrasonic signal to return.

- Trigger pulse is sent from ESP32
- Echo signal is received
- Distance is calculated using time-of-flight principle

4.3.2 Vehicle Count Logic

The system maintains a vehicle count based on entry and exit detection.

- Entry detected → increment count
- Exit detected → decrement count

4.3.3 Slot Detection Logic

Each parking slot is monitored using a threshold distance value.

- Distance $\leq$ threshold $\rightarrow$ Occupied
- Distance $>$ threshold $\rightarrow$ Free

4.4 MQTT Communication

MQTT protocol is used for communication between ESP32 and Node-RED.

Key MQTT Topics:

- parking/status
- parking/slots/filled
- parking/slots/available
- parking/environment/temperature
- parking/environment/humidity
- parking/slot/map

ESP32 acts as a publisher, while Node-RED acts as a subscriber.

4.5 Node-RED Implementation

Node-RED is used to visualize the parking data using a dashboard.

The system uses:

- Text fields for status display
- Gauges for temperature and humidity
- Slot mapping visualization

4.6 Algorithm

1. Initialize ESP32 and sensors
2. Read entry and exit sensor values
3. Update vehicle count
4. Check slot occupancy
5. Read temperature and humidity
6. Publish data via MQTT
7. Update dashboard
8. Repeat continuously

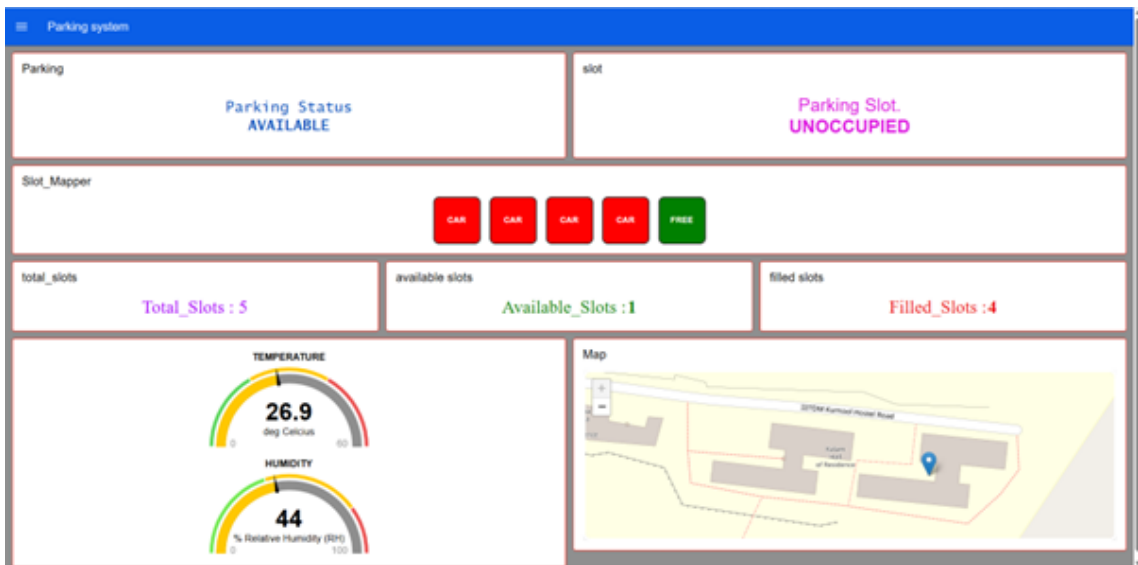


FIGURE 4.3: Smart Parking Dashboard

Chapter 5

Results and Analysis

5.1 Introduction

This chapter presents the results obtained from the implementation of the Smart Parking System. The system was tested under different conditions to evaluate its performance in terms of accuracy, responsiveness, and reliability.

5.2 Serial Monitor Output

The ESP32 serial monitor displays real-time data including vehicle count, slot availability, temperature, humidity, and parking status.

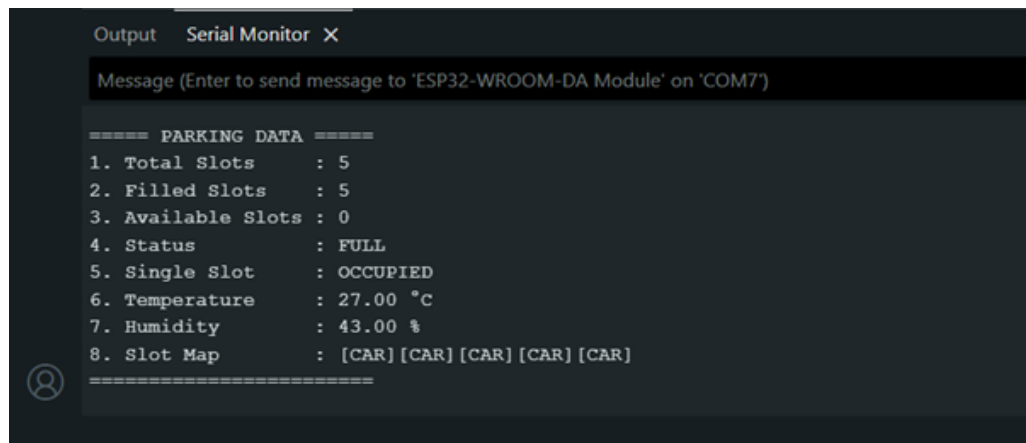


FIGURE 5.1: Serial Monitor Output of Smart Parking System

5.3 Node-RED Dashboard Output

The Node-RED dashboard provides a graphical interface for monitoring parking data.

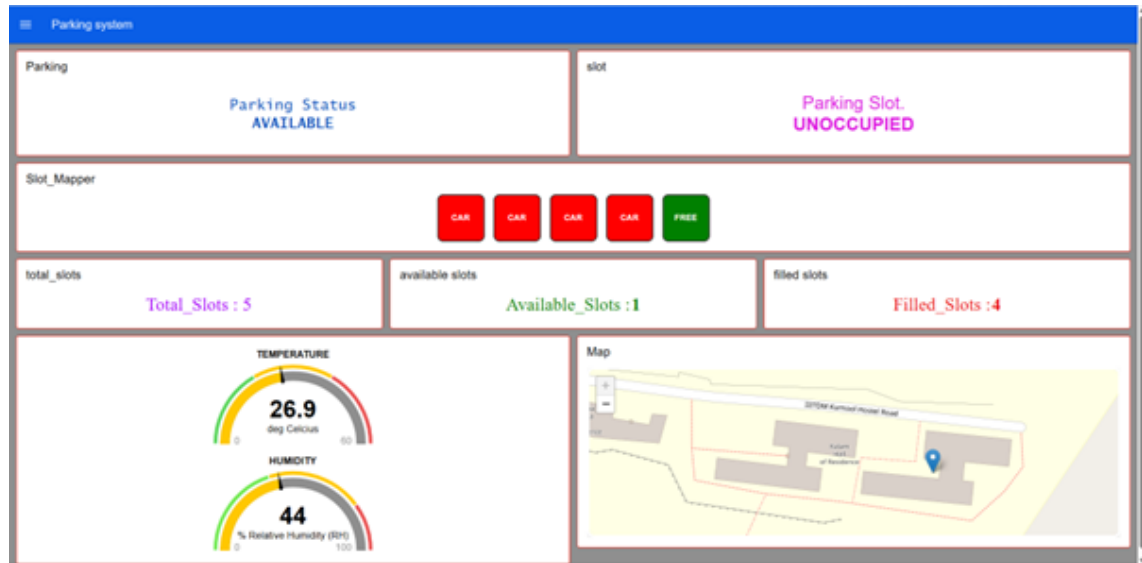


FIGURE 5.2: Node-RED Dashboard Display

The dashboard displays:

- Total parking slots
- Filled slots
- Available slots
- Temperature and humidity
- Slot-wise occupancy

5.4 Mobile Monitoring

The system was also tested using a mobile MQTT client, which allows remote monitoring of parking data in text format.

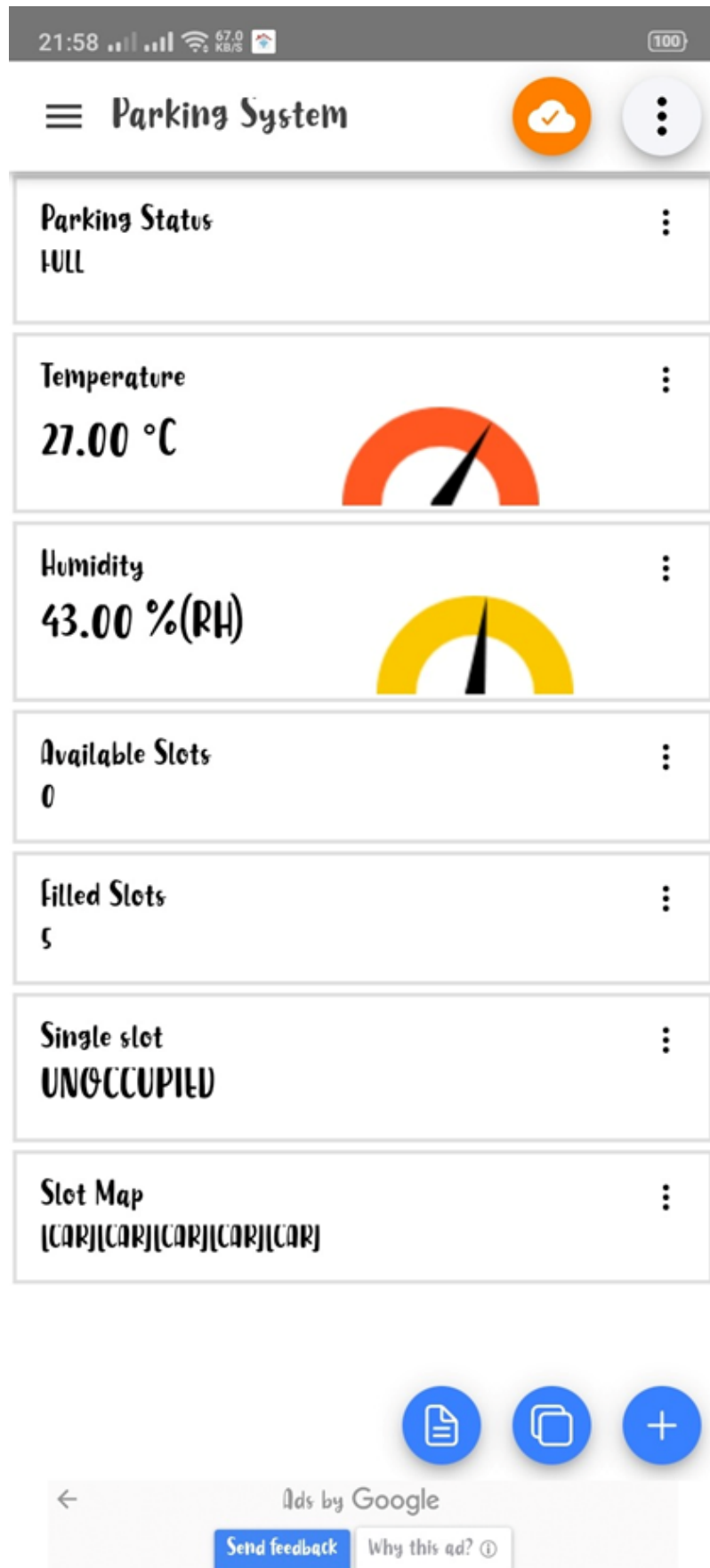


FIGURE 5.3: Mobile Monitoring using MQTT Application

5.5 Performance Analysis

5.5.1 Vehicle Detection Accuracy

Test Case	Actual Count	Detected Count
Test 1	3	3
Test 2	5	5
Test 3	2	2
Test 4	4	4

The system achieved high accuracy in detecting vehicle entry and exit.

5.5.2 Response Time

Operation	Response Time
Sensor Detection	Less than 1 sec
MQTT Transmission	Less than 1 sec
Dashboard Update	Instant

The system demonstrates low latency and real-time performance.

5.5.3 System Reliability

The system was tested continuously for extended durations and showed stable performance without significant errors. The MQTT communication ensured reliable data transmission.

5.6 Discussion

The results show that the Smart Parking System effectively provides real-time parking information with high accuracy and low latency. The integration of sensors, ESP32, MQTT, and Node-RED ensures seamless data flow and efficient monitoring.

The system successfully reduces manual effort and provides a scalable solution for modern parking management.

5.7 Limitations

- Limited to small-scale parking environments
- Sensor accuracy may vary due to environmental conditions
- Requires stable internet connectivity

Chapter 6

Conclusion

The Smart Parking System developed in this project demonstrates an effective and practical implementation of Internet of Things (IoT) technology for real-time parking management. The system successfully integrates ultrasonic sensors, ESP32 microcontroller, MQTT communication protocol, and Node-RED dashboard to monitor parking slot availability and vehicle movement.

The implementation enables automatic detection of vehicle entry and exit, real-time monitoring of parking slots, and remote visualization of data through a user-friendly dashboard. The use of MQTT ensures efficient and reliable communication, while the event-driven data transmission mechanism reduces unnecessary network load and improves system performance.

The results obtained from testing show that the system provides accurate parking information with minimal delay. The integration of hardware and software components was achieved successfully, demonstrating the feasibility of deploying such systems in real-world environments.

Overall, the project highlights the potential of IoT-based solutions in addressing urban challenges such as traffic congestion and inefficient parking management. The developed system is cost-effective, scalable, and suitable for small to medium-scale parking applications.

This project not only strengthened practical knowledge in embedded systems, communication protocols, and IoT platforms but also provided valuable hands-on experience in designing and implementing real-time systems.